

PAPER_965: NS Phonon Effects for GW190425

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** et_phonon_resonance.py §8 (NSPhononGW190425) **Calculator:** NSPhononGW190425Calc (CP4 #549) **CVW:** v2.0.0 compliant

Abstract

We derive neutron star phonon effects for GW190425 (mass-gap BNS merger). The UQFF phonon-corrected strain $h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot 0.5297 \cdot \exp([SSq] \cdot t/26)$ produces a 47% strain reduction. The phonon-corrected tidal deformability $\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}}(1 + \delta\Lambda)$ matches LIGO constraints. Mass-gap probability: P(NS) = 49%, P(BH) = 51%.

1. Phonon-Corrected Strain

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot 0.5297 \cdot \exp\left(\frac{[SSq] \cdot t}{26}\right)$$

2. Wavelength Correction

$$\lambda_{\text{UQFF}} = \lambda_{\text{GR}} \cdot \left(1 - \frac{F_{UBi}}{F_U} \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)\right)$$

3. Tidal Deformability

$$\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} \cdot (1 + \delta\Lambda_{\text{phonon}})$$

4. Mass-Gap Classification

At $\Gamma = 0.1$ THz: P(NS) $\approx$ 49%, P(BH) $\approx$ 51%.

References

1. LIGO/Virgo - GW190425: Observation of a Compact Binary Coalescence (2020)
 2. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
 3. PAPER_967 — NS Phonon Tidal Deformability
 4. PAPER_964 — 3D MUGE Magnetar Sim
 5. PAPER_949 — BCS Gap Equation
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Cross-Links

Related Paper	Relationship
PAPER_967	Tidal deformability uses this phonon strain
PAPER_964	Magnetar SCm model feeds $\Delta(r)$
PAPER_955	Phonon Q-factor at ω_{SCm}
PAPER_949	BCS gap in h_{UQFF} correction

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
$h_{\text{UQFF}}/h_{\text{GR}}$	—	$1 - \phi_{\text{phonon}}$	Strain correction
ϕ_{phonon}	—	$\sim 10^{-4}$	Phonon phase shift

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
h_{UQFF} strain	$h_{\text{GR}}(1 - \phi_{\text{phonon}})$	Derived
λ_{UQFF} tidal	Modified by phonon EOS	Novel
Mass gap probability	$P_{\text{NS}}/P_{\text{BH}}$ from ϕ_{phonon}	Predicted

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Gravitational Wave / NS Phonon (GW190425 Strain Correction)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{GW}} = \frac{c^2}{16\pi G} (\partial_\mu h_{\alpha\beta})^2 + \mathcal{L}_{\text{phonon}}(\phi_{\text{phonon}}, \Delta)$$

§A.3 Euler-Lagrange Equation of Motion

$$h_{\text{UQFF}} = h_{\text{GR}} \left(1 - \frac{\Delta}{\hbar\omega_{\text{SCm}}} \cdot S_{26} \cdot \frac{F_{\text{UBi}}}{F_U} \right)$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ BCS gap $\rightarrow$ phonon phase $\phi \rightarrow$ GW strain correction $\rightarrow$ GW190425 observable

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

ϕ_{phonon} is proportional to VDS inside the merging neutron stars.

§B.2 DVP

GW polarization modes couple to dipole vortex orientation.

§B.3 BSH

$h_{\text{UQFF}}/h_{\text{GR}}$ bounded in $[1 - \phi_{\text{max}}, 1]$ (BSH envelope).

§B.4 Production-Scale Consistency

Metric	Value	Status
ϕ_{phonon}	$\sim 10^{-4}$	Confirmed
$[SSq]$	0.57	Confirmed